NRSG 233 HEALTH ASSESSMENT · MODULE 3

Skin, Hair & Nails
Study guide

For Luke's Module 3 quiz on Tuesday, October 13, 2026, 8:30 am PT.

> **Personal study sheet, not for submission.** NRSG 233 bars AI on submitted work.

Built from: Jarvis & Eckhardt, *Physical Examination & Health Assessment*, 9th ed., Chapter 13 (pp. 199–249) and the pitting-edema scale from Ch. 21 (p. 518), plus the filled Module 3 vocabulary list (31 terms).

Not yet included: Dr. Dunbar's lecture slides (NSRG233SkinPPT.pdf). They'll be added when Canvas access is restored. Her slides win over the textbook if they differ.

Page numbers in [brackets] are Jarvis book pages. Syllabus reading: pp. 111, 207–215, 226–237, 246.

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1. Structure and function

Skin — overview

- Largest organ system; covers about 20 square feet of surface area in the average adult. Guards against environmental stresses (trauma, pathogens, dirt) and adapts to heat and cold. [Jarvis p. 199]
- Two layers: outer **epidermis** and inner supportive **dermis**; beneath these is the **subcutaneous** layer of adipose tissue. [Jarvis p. 199]

Epidermis

- Thin but tough; cells bound tightly into sheets forming a rugged protective barrier. [Jarvis p. 199]
- **Stratum basale** (basal cell layer): inner layer that forms new skin cells; major ingredient is the tough fibrous protein **keratin**. [Jarvis p. 199]

- **Melanocytes** interspersed in this layer produce pigment **melanin** (brown tones of skin and hair). All skin colors have the same number of melanocytes; the amount of melanin produced varies with genetic, hormonal, and environmental influences. [Jarvis p. 199]
- New cells migrate up and flatten into the outer **stratum corneum** (horny cell layer) — dead keratinized cells, interwoven and closely packed; constantly shed (**desquamated**) and replaced from below. Epidermis is completely replaced every 4 weeks. [Jarvis p. 199]
- Thicker on palms and soles because of work and weight bearing. Epidermis is **avascular**, nourished by dermal blood vessels below. [Jarvis p. 200]

> Skin color comes from three sources: (1) mainly brown melanin, (2) yellow-orange carotene, (3) red-purple blood in the underlying vascular bed. All people have skin with varying shades of brown, yellow, and red. Color is further modified by skin thickness and the presence of edema. [Jarvis p. 200]

Dermis

- Inner supportive layer, mostly connective tissue (**collagen**) — tough, fibrous protein that resists tearing. [Jarvis p. 200]
- Contains resilient elastic tissue allowing stretch with body movement; also holds nerves, sensory receptors, blood vessels, and lymphatics. [Jarvis p. 200]
- Appendages from the epidermis — hair follicles, sebaceous glands, sweat glands — are embedded in the dermis. [Jarvis p. 200]

Subcutaneous layer

- Adipose tissue — lobules of fat cells. Stores fat for energy, insulates for temperature control, cushions for protection. [Jarvis p. 200]
- Loose layer gives skin increased mobility over structures underneath. [Jarvis p. 200]

Hair

- Threads of keratin. **Hair shaft** = visible projecting part; **root** = below the surface in the follicle. At the root, the **bulb matrix** is the expanded area where new cells are produced at a high rate. [Jarvis p. 200]
- Growth is cyclical (active and resting phases); each follicle functions independently. [Jarvis p. 200]
- **Arrector pili** muscles surround the follicle; contraction elevates the hair ("goose flesh") with cold or emotion. [Jarvis p. 200]
- **Vellus hair** — fine, faint; covers most of the body except palms, soles, dorsa of distal fingers, umbilicus, glans penis, and inside the labia. [Jarvis p. 200]
- **Terminal hair** — darker, thicker; on scalp and eyebrows, and after puberty on axillae, pubic area, and (male) face and chest. [Jarvis p. 200]

Sebaceous glands

- Produce **sebum**, a protective lipid secreted through the hair follicles; oils and lubricates skin and hair and forms an emulsion with water that retards water loss. [Jarvis p. 200]
- Dry skin results from loss of water, not directly from loss of oil. [Jarvis p. 200]
- Present everywhere except palms and soles; most abundant in scalp, forehead, face, and chin. [Jarvis p. 200]

Sweat glands

- **Eccrine** — coiled tubules opening directly onto the skin surface; produce dilute saline sweat; evaporation reduces body temperature. Widely distributed; mature in the 2-month-old infant. [Jarvis p. 200]
- **Apocrine** — produce a thick, milky secretion; open into hair follicles; located mainly in axillae, anogenital area, nipples, and navel; vestigial in humans. Active at puberty; secretion with emotional and sexual stimulation. Skin flora react with apocrine sweat to produce musky body odor. Function decreases in the aging adult. [Jarvis p. 200]

Nails

- Hard plates of keratin on the dorsal edges of fingers and toes. **Nail plate** is clear with fine longitudinal ridges that become prominent in aging. [Jarvis p. 200]
- Take pink color from the underlying vascular **nail bed**. **Lunula** = white, opaque, semilunar area at the proximal end, over the nail matrix where new keratinized cells form. [Jarvis p. 200]
- **Nail folds** overlap posterior and lateral borders; the **cuticle** works like a gasket to cover and protect the nail matrix. [Jarvis p. 200]

Functions of the skin

- Protection — minimizes injury from physical, chemical, thermal, and light-wave sources. [Jarvis p. 200]
- Prevents penetration — defense barrier against microorganisms and loss of water and electrolytes. [Jarvis p. 200]
- Perception — sensory surface with neurosensory end-organs for touch, pain, temperature, pressure. [Jarvis p. 201]
- Fluid balance — about 500 mL fluid lost daily by insensible perspiration, plus visible sweat. [Jarvis p. 201]
- Temperature regulation — heat dissipation through sweat glands; heat storage through subcutaneous insulation. [Jarvis p. 201]
- Identification — unique combinations of facial characteristics, hair, skin color, fingerprints. [Jarvis p. 201]
- Communication — emotions via face, body posture, and vascular mechanisms such as blushing or blanching. [Jarvis p. 201]
- Wound repair — cell replacement of surface wounds. [Jarvis p. 201]
- Absorption and excretion — limited excretion of metabolic wastes (minerals, sugars, amino acids, cholesterol, uric acid, urea). [Jarvis p. 201]
- Production of vitamin D — UV light converts cholesterol into vitamin D at the skin surface. [Jarvis p. 201]

2. Health history questions

Examiner asks	Why it matters
Past history of skin disease (allergies, hives, psoriasis, eczema)? How treated? Family history of allergies? Known allergies to drugs, plants, animals? Birthmarks, tattoos? [Jarvis p. 203]	Significant family predisposition: allergies, hay fever, psoriasis, atopic dermatitis, acne. Identify offending allergen. Tattoos: atypical mycobacterial infections, inflammatory and hypersensitivity reactions can occur. [Jarvis p. 203]
Change in pigmentation — generalized or localized? [Jarvis p. 203]	Hypo- or hyperpigmentation. Generalized change suggests systemic illness: pallor, jaundice, cyanosis. [Jarvis p. 203]
Change in a mole — color, size, shape, sudden tenderness, bleeding, itching? Any "sores" that do not heal? [Jarvis p. 203]	Signs suggest neoplasm in pigmented nevus. Patient may not see changes on back or buttocks. [Jarvis p. 203]
Change in feel of skin: temperature, moisture, texture? Excess dryness — seasonal or constant? [Jarvis p. 203]	Seborrhea — oily; xerosis — dry. [Jarvis p. 203]
Pruritus — mild (prickling, tingling) or intense? Awaken you from sleep? Where, when did it start? Other skin pain or soreness? [Jarvis p. 203]	Pruritus is the most common skin symptom; occurs with dry skin, aging, drug reactions, allergy, obstructive jaundice, uremia, lice. Presence/absence aids diagnosis; scratching causes excoriation of the primary lesion. [Jarvis p. 203]

Examiner asks	Why it matters
Excess bruising? Where? How did it happen? How long? [Jarvis p. 203]	Multiple cuts and bruises, bruises in various stages of healing, bruises above knees and elbows, illogical explanation — consider physical abuse. Frequent falls may stem from neurologic or cardiovascular dizziness; minor trauma may reflect aging skin or alcohol/drug abuse. [Jarvis p. 203]
Rash or lesion — onset, location, where it spread, color, raised or flat, crust, odor, tender, warm? Duration? Setting: similar rash at home/work, camping, new pet, new food or drug, stress? Alleviating/aggravating factors (home remedies, bath, lotions, heat)? Associated itching or fever? What do you think it means? Coping strategies — effect on self-care, hygiene, work/home/social function? New or increased stress? [Jarvis pp. 203-204]	Rashes are a common reason for seeking care; history may predict lesion type and cause. Identify primary site, migration pattern, evolution, new exposures or contacts. Myriad OTC remedies are tried first. Assess perception of cause (fear of cancer, tickborne illness, STI). Chronic skin disease may cause poor self-esteem, social isolation, anxiety; stress can exacerbate chronic skin illness. [Jarvis pp. 203-204]
Medications — prescription and OTC? Recent change? How long on medication? [Jarvis p. 204]	Drugs (especially antibiotics) may cause allergic skin eruption. Increased sunlight sensitivity: sulfonamides, thiazide diuretics, oral hypoglycemic agents, tetracycline. Hyperpigmentation: antimalarials, anticancer agents, hormones, metals, tetracycline. Sensitivity may develop even after long use. [Jarvis p. 204]

Examiner asks	Why it matters
Hair loss — gradual or sudden? Symmetric? With fever, illness, increased stress? Unusual hair growth? Change in texture or appearance? [Jarvis p. 204]	Alopecia is significant loss; a full head of hair equates with vitality in many cultures. If treated trivially, the person may seek unproven treatments. Hirsutism = shaggy or excessive hair. [Jarvis p. 204]
Change in nails — shape, color, brittleness? Nail biting or chewing? [Jarvis p. 204]	To cut disease transmission, keep fingers out of mouth. [Jarvis p. 204]
Environmental/occupational hazards — dyes, toxic chemicals, radiation? Hobbies, household or furniture repair? Sun exposure from work, leisure, sunbathing, tanning salons? Insect bites (bee, tick, mosquito)? Recent exposure to plants, animals in yard work or camping? [Jarvis p. 204]	Majority of skin cancers result from environmental or occupational agents. At risk: outdoor workers and sports enthusiasts, farmers, sailors, sun seekers, creosote workers, roofers, coal workers. Unprotected sun exposure accelerates aging and produces lesions; more at risk: light-skinned people, light eye and hair color, freckles, those regularly in sun. Identify contactants causing contact dermatitis; keep a diary for chronic recurrent urticaria. [Jarvis p. 204]
Patient-centered care — what do you do to care for skin, hair, nails? Cosmetics, soaps, chemicals? Cuticle clipping, adhesive for false fingernails? How do you control your environment if allergic? Do you perform skin self-examination? [Jarvis pp. 204-205]	Assess self-care and influence on self-concept (media emphasis on beauty norms). Many OTC remedies are costly and exacerbate skin problems. See Patient Teaching for self-examination. [Jarvis pp. 204-205]

Additional history — infants and children

- Birthmarks? Any change in skin color as a newborn — jaundice (which day after birth?) or cyanosis (circumstances?) [Jarvis p. 205]

- Rash or sores — what brings it on? New food or formula? Chocolate, cow's milk, eggs? Generalized rash suggests allergic reaction; irritability/fussiness may indicate pruritus. [Jarvis p. 205]
- Diaper rash — care, washing of diapers, changing frequency, skin cleansing. Occlusive diapers or infrequent changing may cause rash; allergy to detergent or wipes possible. [Jarvis p. 205]
- Burns or bruises — where, how did it happen? Cigarette burns, excessive bruising above knees or elbows, linear whip marks suggest abuse; history often does not coincide with the physical appearance. [Jarvis p. 205]
- Exposure to contagious conditions (scabies, impetigo, lice), communicable diseases (measles, chickenpox, scarlet fever), or toxic plants (poison ivy)? Vaccinations up to date? [Jarvis p. 205]
- Habits — nail-biting, twisting hair, rubbing head on mattress? [Jarvis p. 205]
- Sun protection — sunscreens, sunblocks, treatment of sunburn. Excessive sun, including severe or blistering sunburns in childhood, increases risk for melanoma in later life. [Jarvis p. 205]

Additional history — adolescent

- Skin problems such as pimples, blackheads? How long? How treated? How do you feel about it? Over 85% of teenagers have acne; psychological effect is significant (poor self-esteem, scarring, depression). Multifactorial causes: increased sebum production, microbes, inflammation. [Jarvis p. 205]

Additional history — aging adult

- Changes noticed in skin over the past few years? Assess impact of aging on self-concept; many "aging" changes, including skin cancers, result from chronic sun damage. [Jarvis p. 205]
- Delay in wound healing? Skin itching? Senile pruritus is usually caused by dry skin (xerosis), too-frequent bathing, or some soaps; pruritus may also reflect drug side effects or systemic disease (liver or kidney disease, cancer, lymphoma). [Jarvis p. 206]
- Other skin pain? Herpes zoster produces intense pain and itching; diabetes may reduce pain sensation in extremities; some older adults accept chronic pain as "part of growing old." [Jarvis p. 206]
- Changes in feet, toenails? Bunions? Possible to wear shoes? Some cannot reach the feet for self-care. [Jarvis p. 206]

- Falls this year — how many? Multiple bruises, trauma from falls. [Jarvis p. 206]
- History of diabetes or peripheral vascular disease? Risk for skin lesions in feet or ankles. [Jarvis p. 206]
- Skin care practices? Bland lotion retains moisture in aging skin; cosmetics, creams, ointments, dyes may cause dermatitis. Aging skin has a delayed inflammatory response to irritants. [Jarvis p. 206]

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3. Assessment technique: inspection and palpation

Preparation

- Control external variables that change skin color (e.g., cold, pale hands from a chilly room). [Jarvis p. 206]
- Know the person's normal skin coloring — baseline knowledge is essential; if first exam, ask about usual skin color and self-monitoring practices. [Jarvis p. 206]
- Skin assessment is integrated throughout the complete examination. Starting with hands and fingernails is a nonthreatening way to become accustomed to your touch; scrutinize the outer skin surface before the underlying structures. [Jarvis p. 206]
- Separate intertriginous areas (skinfolds — under large breasts, obese abdomen, groin); these are dark, warm, moist and prone to irritation or infection. Always remove socks and inspect feet, toenails, and folds between toes. [Jarvis pp. 206-207]
- Regional exam: help the person remove clothing; stand back for an overall impression to reveal distribution patterns, then inspect lesions carefully. With a rash, check all areas of the body and inspect mucous membranes. [Jarvis p. 207]

Equipment

- Strong direct lighting (natural daylight ideal; halogen light will suffice), small centimeter ruler, penlight, gloves. [Jarvis p. 206]
- For special procedures: Wood's light (filtered UV light), dermatoscope. [Jarvis p. 206]

Color

- Observe skin tone — normally even and consistent with genetic background; varies from pinkish tan to ruddy dark tan or light to dark brown, with possible yellow or olive overtones. Dark-skinned people normally have lighter pigmentation on palms, nail beds, and lips. [Jarvis p. 207]
- General pigmentation is darker in sun-exposed areas. Benign pigmented areas: freckles (ephelides — small flat macules of brown melanin on sun-exposed skin); mole (nevus — clump of melanocytes, tan to brown, flat or raised; acquired nevi are symmetric, ≤6 mm, smooth borders, uniform pigmentation); birthmarks (may be tan to brown). [Jarvis pp. 207-208]
- For widespread color change: in dark-skinned people normal pigment may mask changes; lips and nail beds may not be accurate signs. More reliable sites have the least pigmentation — under the tongue, buccal mucosa, palpebral conjunctiva, sclera. [Jarvis p. 208]
- Pallor: check mucous membranes, lips, nail beds; for anemia, check palpebral conjunctiva and nail beds near outer and inner canthi (coloration often lighter near inner canthus). In dark skin look for absence of luster; brown skin appears yellowish-brown, Black skin ashen or gray. [Jarvis p. 208]
- Erythema/fever/inflammation in dark skin: you cannot see inflammation, so palpate for increased warmth, taut or tightly pulled surfaces indicating edema, and hardening of deep tissues or blood vessels. [Jarvis p. 208]
- Cyanosis is best seen in lips, nose, cheeks, ears, and oral mucous membranes and in artificial fluorescent light; do not confuse with the normal bluish tone on lips of dark-skinned persons of Mediterranean origin. [Jarvis p. 208]
- Jaundice is first noted at the junction of the hard and soft palate and in the sclera; best assessed in direct natural daylight. Do not mistake normal yellow subconjunctival fatty deposits or yellow calluses on palms and soles for jaundice. [Jarvis p. 209]

Temperature, moisture, texture, thickness

- Palpate the skin: should be warm and equal bilaterally; warmth suggests normal circulatory status. Hands and feet may be slightly cooler in a cool environment. [Jarvis p. 209]

- Moisture: perspiration normally appears on face, hands, axillae, and skinfolds with activity, warm environment, or anxiety. Assess dehydration in the oral mucous membranes (normally smooth and moist). Dark skin may normally look dry and flaky without systemic dehydration. [Jarvis p. 210]
- Texture: normal skin feels smooth and firm with an even surface. [Jarvis p. 210]
- Thickness: epidermis is uniformly thin over most of the body; thickened callus areas on palms and soles are normal (circumscribed overgrowth of epidermis from pressure and friction). [Jarvis p. 210]

Edema

- Edema = fluid accumulating in interstitial spaces; not present normally. Imprint your thumbs firmly for 3 to 4 seconds against the ankle malleolus or the tibia, then lift. Normally the skin surface stays smooth; if pressure leaves a dent, "pitting" edema is present. [Jarvis p. 210]
- Pitting edema scale (Jarvis Ch. 21) [Jarvis p. 518]:

Grade	What you see
1+	Mild pitting, slight indentation, no perceptible swelling of the leg
2+	Moderate pitting, indentation subsides rapidly
3+	Deep pitting, indentation remains for a short time, leg looks swollen
4+	Very deep pitting, indentation lasts a long time, leg is grossly swollen and distorted

> Jarvis notes this scale is subjective (not proven reliable); ankle circumference with a nonstretchable tape is more reliable. Jarvis gives no mm depths. If her slides use mm (2/4/6/8 mm), learn hers. [Jarvis pp. 518–519]

- Edema masks normal skin color and obscures jaundice or cyanosis; it makes dark skin look lighter. It shows in dependent parts (feet, ankles, sacral areas) where skin looks puffy and tight. [Jarvis p. 210]

Mobility and turgor

- Pinch up a large fold of skin on the anterior chest under the clavicle. **Mobility** = ease of skin to rise; **turgor** = ability to return to place promptly when released (reflects elasticity). [Jarvis p. 210]

Vascularity and bruising

- Cherry (senile) angiomas — small (1 to 5 mm), smooth, slightly raised bright red dots on the trunk in all adults older than 30 years; increase in size and number with aging; not significant. [Jarvis p. 211]
- Any bruising (contusion) should be consistent with expected trauma of life; normally no venous dilations or varicosities. [Jarvis p. 211]
- Document tattoos; inspect for infection or inflammation (normally no reactions). [Jarvis p. 211]

Lesions — what to note

Characteristic	What to note
Color	Document the lesion's color. [Jarvis p. 211]
Elevation	Flat, raised, or pedunculated. [Jarvis p. 211]
Pattern or shape	Grouping or distinctness — e.g., annular, grouped, confluent, linear; pattern may be characteristic of a certain disease. [Jarvis p. 211]
Size	Measure in centimeters with a ruler; avoid "quarter size" or "pea size." [Jarvis p. 211]
Location and distribution	Generalized or localized to a specific irritant; around jewelry, watchband, eyes? [Jarvis p. 211]
Exudate	Presence, color, any odor. [Jarvis p. 211]

- **Primary lesion** — develops on previously unaltered skin. **Secondary lesion** — a primary lesion changed by time, scratching, or infection. [Jarvis p. 211]

- Palpate lesions (glove if contact with blood, mucosa, or body fluid is expected). Roll a nodule between thumb and index finger to assess depth; gently scrape a scale with a tongue blade to see if it comes off; note the nature of its base and whether it bleeds; note surrounding skin temperature. Erythema with rashes is not always accompanied by increased skin temperature. [Jarvis p. 212]
- Blanch test: stretch the skin between thumb and index finger to decrease normal red tones and brighten macules. Red macules from dilated blood vessels blanch momentarily; those from extravasated blood (petechiae) do not. Blanching also helps identify a macular rash in dark-skinned people. [Jarvis p. 212]
- **Dermatoscope** — blocks skin surface reflections for closer inspection; aids early detection of melanoma, scabies, wart versus foreign body, fungal infections. For experienced examiners it helps screen benign from cancerous lesions; the light beam falls at 20°, eliminating light reflection. [Jarvis p. 212]
- **Wood's light** (UV light filtered through special glass) — with room darkened, shine on the area to detect fluorescing lesions; blue-green fluorescence indicates fungal infection (e.g., tinea capitis). [Jarvis p. 212]

> Common shapes/configurations of lesions: annular (circular, center to periphery — tinea corporis, tinea versicolor, pityriasis rosea); confluent (lesions run together — urticaria); discrete (separate individual lesions — skin tags, acne); gyrate (twisted, coiled spiral, snakelike); grouped (clusters — vesicles of contact dermatitis); linear (scratch, streak, line, stripe); target/iris (concentric rings of color — erythema multiforme); zosteriform (linear along a unilateral nerve route — herpes zoster); polycyclic (annular lesions grow together — lichen planus, psoriasis). [Jarvis pp. 228-229]

> **Primary lesions:** macule (flat color change <1 cm); patch (macules >1 cm); papule (solid, elevated, circumscribed, <1 cm); plaque (papules coalescing, surface elevation wider than 1 cm, plateaulike); nodule (solid, elevated, hard or soft, >1 cm, may extend deeper); tumor (larger than a few centimeters, firm or soft, deeper into dermis); wheal (superficial, raised, transient, erythematous, irregular from edema); urticaria (wheals coalesce, intensely pruritic); vesicle (elevated cavity with free

> fluid up to 1 cm — a "blister"); bulla (larger than 1 cm, usually unilocular, thin-walled); cyst (encapsulated fluid-filled cavity in dermis or subcutaneous layer); pustule (turbid fluid/pus, circumscribed and elevated). [Jarvis pp. 230-231]
> **Secondary lesions:** crust (dried exudate); scale (compact desiccated flakes); fissure (linear crack extending into dermis); erosion (shallow, epidermis lost, heals without scar); ulcer (deeper, into dermis, may bleed, leaves scar); excoriation (self-inflicted abrasion from scratching); scar (permanent fibrotic change); atrophic scar (depressed, tissue loss — e.g., striae); lichenification (thickened skin from prolonged scratching); keloid (benign excess scar tissue beyond original injury). [Jarvis pp. 231-233]

Hair and nails — technique points

- Hair: separate into sections and lift, observing the scalp; with itching, inspect behind the ears and occipital area. Loose white flakes = seborrhea (dandruff), common and normal. [Jarvis p. 213]
- Nail shape/contour: surface normally slightly curved or flat; posterior and lateral nail folds smooth and rounded; edges smooth, rounded, clean. [Jarvis p. 213]
- **Profile sign:** view the index finger in profile; the angle of the nail base should be about 160 degrees and the base firm to palpation. Curved nails are a normal variation with a convex profile but a normal angle (160 degrees or less). [Jarvis p. 214]
- Consistency: surface smooth and regular, not brittle or splitting; thickness uniform; nail firmly adherent to the nail bed. [Jarvis p. 214]
- Color: translucent nail plate shows an even pink nail bed. White hairline linear markings (leukonychia) are normal from trauma or picking. Dark-skinned people may have brown-black pigmented areas or linear bands along the nail edge. [Jarvis p. 215]
- **Capillary refill:** with index or middle fingertip at heart level, depress the nail edge at least 5 seconds to blanch, then release. Color return is normally instant or within a few seconds in a cold environment; a healthy return takes 1 or 2 seconds. [Jarvis p. 215]
- Inspect toenails; separate the toes and note the smooth skin in between. [Jarvis p. 215]

4. Normal vs abnormal findings, light vs dark skin

Table 13.1 — Detecting color changes in light and dark skin

Change	Light skin	Dark skin	Cause
Pallor (anemia, shock)	Generalized pallor	Brown skin is yellow-brown, dull; Black skin is ashen gray, dull; skin loses its healthy glow	Decreased hematocrit; decreased perfusion and vasoconstriction. Check areas with least pigmentation (conjunctivae, mucous membranes). [Jarvis p. 226]
Pallor — acute arterial insufficiency	Sudden onset, marked localized pallor or mottling (e.g., lower extremities, especially when elevated), absent pulse(s)	Ashen gray, dull; cool to palpation; sudden onset; absent pulse(es)	Acute arterial insufficiency. [Jarvis p. 226]
Pallor — chronic arterial insufficiency	Pallor on elevation, rubor on dependency	Inspect for swelling, edema	Chronic arterial insufficiency. [Jarvis p. 226]
Albinism	Whitish pink	Tan, cream, white	Total absence of pigment melanin throughout the integument. [Jarvis p. 226]
Vitiligo	Patchy milky-white spots, often symmetric bilaterally	Same	Patchy depigmentation from destruction of melanocytes. [Jarvis p. 226]

Change	Light skin	Dark skin	Cause
Cyanosis	Dusky blue; peripheral — nail beds dusky	Dark but dull, lifeless; only severe cyanosis is apparent in skin. Check conjunctivae, oral mucosa, nail beds	Increased unoxygenated hemoglobin — central (chronic heart and lung disease with arterial desaturation) or peripheral (exposure to cold, anxiety). [Jarvis p. 226]
Erythema	Red, bright pink	Purplish tinge but difficult to see; palpate for increased warmth with inflammation, taut skin, hardening of deep tissues	Hyperemia — increased blood in engorged arterioles (inflammation, fever, alcohol intake, blushing). [Jarvis p. 227]
Erythema — polycythemia	Ruddy blue in face, oral mucosa, conjunctiva, hands, feet	Well concealed by pigment; check for redness in lips	Increased red blood cells, capillary stasis. [Jarvis p. 227]
Erythema — carbon monoxide poisoning	Bright cherry red in face and upper torso	Cherry-red color in nail beds, lips, oral mucosa	Carbon monoxide poisoning. [Jarvis p. 227]
Erythema — venous stasis	Dusky rubor of dependent extremities; a prelude to necrosis with pressure sore	Easily masked; use palpation for warmth or edema	Decreased blood flow from area, engorged venules. [Jarvis p. 227]

Change	Light skin	Dark skin	Cause
Jaundice	Yellow in sclera, hard palate, mucous membranes, then over skin	Check sclera for yellow all the way up to iris (do not mistake normal yellowish fatty deposits in scleral periphery); best noted at junction of hard and soft palate, also palms	Increased serum bilirubin from liver inflammation or hemolytic disease (e.g., after severe burns, some infections). [Jarvis p. 227]
Carotenemia	Yellow-orange in forehead, palms and soles, nasolabial folds — but no yellowing in sclera or mucous membranes	Yellow-orange tinge in palms and soles	Increased deposit of carotene from high intake of carrots, yellow squash, even baby food. [Jarvis p. 227]
Uremia	Orange-green or gray overlying pallor of anemia; may also have ecchymoses and purpura	Easily masked; rely on laboratory and clinical findings	Renal failure causes retained urochrome pigments in the blood. [Jarvis p. 228]
Brown-tan — Addison disease	Bronzed appearance, an "eternal tan," most apparent around nipples, perineum, genitalia, pressure points (inner thighs, buttocks, elbow, axillae)	Easily masked; rely on laboratory and clinical findings	Cortisol deficiency stimulates increased melanin production. [Jarvis p. 228]

Change	Light skin	Dark skin	Cause
Café au lait spots	Tan to light brown, irregularly shaped, oval patch with well-defined borders	Same	Increased melanin pigment in basal cell layer. [Jarvis p. 228]
Petechiae	Tiny punctate hemorrhages, 1 to 3 mm, round, discrete; dark red, purple, or brown; will not blanch	Best visualized in areas of lighter melanization (abdomen, buttocks, volar forearm); when skin is very dark, petechiae cannot be seen	Bleeding from superficial capillaries; may indicate abnormal clotting factors. Inspect mouth (buccal mucosa) and conjunctivae. [Jarvis p. 237]
Ecchymosis	Purplish patch from extravasation of blood into the skin, >3 mm in diameter	Difficult to see in dark-skinned people	Extravascular blood from breaks in vessels; pressure will not cause blanching. Note: a bruise (contusion) differs — it is caused by blunt force trauma. [Jarvis pp. 236-237]

Other normal vs abnormal points

- **Vitiligo** — acquired complete absence of melanin in patchy areas of white or light skin on face, neck, hands, feet, body folds, around orifices. Occurs in all people; dark-skinned people are more severely affected with greater threat to body image. [Jarvis p. 207]
- **Carotenemia** (infants) — yellow-orange color in light-skinned persons with no yellowing of sclera or mucous membranes; from large amounts of carotene-containing foods; best seen on palms, soles, forehead, tip of nose, nasolabial folds, chin, behind ears, over knuckles; fades within 2 to 6 weeks of withdrawing carotene-rich foods. [Jarvis p. 218]
- **Cyanosis cautions** — an anemic person may have hypoxemia without looking blue; a person with polycythemia looks ruddy blue

at all times and may not be hypoxemic. Cyanosis is difficult to observe in darkly pigmented people; rely on level of consciousness changes and respiratory distress signs. [Jarvis p. 209]

- **Temperature** — generalized coolness with shock, cardiac arrest; localized coolness with peripheral arterial insufficiency and Raynaud disease. Generalized hyperthermia with fever or heavy exercise; localized hyperthermia with trauma, infection, or sunburn. Hyperthyroidism: warm, moist skin; chronic cigarette smoking: cool, pale hands. [Jarvis pp. 209-210]
- **Moisture** — diaphoresis with thyrotoxicosis, heart attack, anxiety, or pain. Dehydration: dry mucous membranes, parched cracked lips; extreme dryness = fissured skin resembling cracks in a dry lake bed. [Jarvis p. 210]
- **Texture** — hyperthyroidism: skin smoother and softer like velvet; hypothyroidism: rough, dry, flaky. [Jarvis p. 210]
- **Thickness** — very thin, shiny (atrophic) skin with arterial insufficiency. [Jarvis p. 210]
- **Edema** — unilateral edema has a local or peripheral cause; bilateral or generalized edema (anasarca) suggests a central problem such as heart failure or kidney failure. [Jarvis p. 210]
- **Mobility/turgor** — decreased mobility with edema; poor turgor in severe dehydration or extreme weight loss ("tents" and stands by itself); scleroderma = decreased mobility. [Jarvis p. 210]
- **Bruising** — multiple bruises at different stages of healing and excessive bruises above knees or elbows raise concern about physical abuse; needle marks may appear on antecubital fossae, forearms, or any available vein. [Jarvis p. 211]
- **Pigmented lesion danger signs (ABCDEF)** — Asymmetry; Border irregularity (notching, scalloping, ragged edges, poorly defined margins); Color variation (brown, tan, black, blue, red, white, or combination); Diameter greater than 6 mm (size of a pencil eraser), although early melanomas may be smaller; Elevation or Evolution (sudden appearance or recent change); Funny looking ("ugly duckling" sign). Additional symptoms: rapidly changing lesion, new pigmented lesion, itching, burning, or bleeding in a mole — warrant referral. [Jarvis pp. 207-208]
- **Nails** — clubbing with congenital cyanotic heart disease, lung cancer, pulmonary diseases (early clubbing: angle straightens to 180 degrees and nail base feels spongy). "Spoon" nails with

chronic iron deficiency anemia. Pits, transverse grooves, or lines may indicate nutrient deficiency or acute illness. Nails thickened and ridged with arterial insufficiency. Suspicious darkly pigmented nail bands with varied pigmentation and width >3 mm in light-skinned persons suggest melanoma. Sluggish color return >3 seconds indicates clinical decline (cardiovascular or respiratory failure, septic shock). [Jarvis pp. 208, 214-215]

- **Hair** — dull, coarse, or brittle scalp hair is abnormal; gray, scaly, well-defined areas with broken hairs accompany tinea capitis. Loss of eyebrows and scalp hair is expected with chemotherapy and hypothyroidism. Absent or sparse genital hair suggests endocrine abnormalities; hirsutism (excess body hair in females in a male pattern) indicates increased androgen. [Jarvis p. 213]
- **Dark skin — common variations** — keloids (raised scars growing beyond wound boundaries), postinflammatory hypopigmentation or hyperpigmentation after acne, pseudofolliculitis ("razor bumps"/ingrown hairs) from close shaving, and melasma (patchy tan-to-dark brown facial discoloration, the "mask of pregnancy"). Hair of persons of African descent has an elliptical hair shaft with asymmetric follicle position and is more susceptible to damage from chronic traction or trauma. [Jarvis p. 202]
- **Aging adult** — solar lentigines (small flat brown macules from chronic sun exposure); seborrheic keratosis (dark, greasy, "stuck on," not malignant); actinic keratosis (red-tan scaly plaques on sun-exposed surfaces, the most frequent premalignant skin lesion in White persons, may develop into squamous cell carcinoma); xerosis (dry, itchy, flaky skin); acrochordons (skin tags); sebaceous hyperplasia (raised yellow papules with central depression); parchment-thin skin with diminished subcutaneous fat and increased risk of pressure injuries and skin tears. Senile purpura — dark red discoloration from minor trauma due to vascular fragility. [Jarvis pp. 201, 221-222]
- **Infants** — mongolian spot (blue-black-to-purple macular area at sacrum/buttocks; present in 90% of Black persons, 80% of Asian and Native American persons, 9% of White persons; do not confuse with bruising). Café au lait macules: six or more, each >1.5 cm in diameter, are diagnostic of neurofibromatosis. Normal erythematous states: beefy red flush of the first 24 hours, harlequin color change, erythema toxicum (flea bite rash) in the first 3 to 4 days. Temporary cyanotic states: acrocyanosis and cutis marmorata (both disappear with warming). Physiologic

jaundice appears after the 3rd or 4th day in about half of newborns; jaundice on the 1st day may indicate hemolytic disease, after 2 weeks may indicate biliary tract obstruction. Milia = tiny white papules from sebum occluding follicles; resolve spontaneously. [Jarvis pp. 216-218]

- **Pregnancy** — striae (jagged linear silver-to-pink "stretch marks," 2nd trimester, in half of all pregnancies); linea nigra (brownish-black midline line); chloasma (irregular brown facial patch); vascular spiders (spider angioma) from increased estrogen — more than 5 occur with significant liver disease. [Jarvis p. 220]

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5. Primary, secondary and vascular lesions

Lesions are traumatic or pathologic changes in previously normal structures. **Primary** = develops on previously unaltered skin; **secondary** = changes over time or from scratching/infection. Note color, elevation, pattern/shape, size in cm (use a ruler, not "pea size"), location/distribution, and any exudate. [Jarvis p. 211]

Shapes and configurations of lesions (Table 13.2)

Shape	Pattern	Example
Annular	Circular; begins in center, spreads to periphery	Tinea corporis (ringworm), tinea versicolor, pityriasis rosea
Confluent	Lesions run together	Urticaria (hives)
Discrete	Distinct, individual lesions remaining separate	Skin tags (acrochordon), acne
Gyrate	Twisted, coiled spiral, snakelike	—
Grouped	Clusters of lesions	Vesicles of contact dermatitis
Linear	Scratch, streak, line, or stripe	—
Target (iris)	Resembles iris of eye; concentric rings of color	Erythema multiforme
Zosteriform	Linear along a unilateral nerve route	Herpes zoster

Shape	Pattern	Example
Polycyclic	Annular lesions grow together	Lichen planus, psoriasis

[Jarvis pp. 228–229]

Primary skin lesions (Table 13.3)

Immediate results of a specific causative factor on previously unaltered skin. [Jarvis p. 230]

Lesion	Description / size	Example
Macule	Flat, circumscribed color change only; <1 cm	Freckles, flat nevi, hypopigmentation, petechiae, measles, scarlet fever
Patch	Macules larger than 1 cm	Mongolian spot, vitiligo, café au lait spot, chloasma, measles rash
Papule	Solid, elevated, circumscribed; superficial epidermal thickening; <1 cm	Elevated nevus (mole), lichen planus, molluscum, wart (verruca)
Plaque	Papules coalesce; plateau-like, disk-shaped elevation >1 cm	Psoriasis, lichen planus
Nodule	Solid, elevated, hard or soft; >1 cm; may extend deeper into dermis than a papule	Xanthoma, fibroma, intradermal nevi
Tumor	Larger than a few centimeters; firm or soft; deeper into dermis; benign or malignant	Lipoma, hemangioma
Wheal	Superficial, raised, transient, erythematous; slightly irregular from edema	Mosquito bite, allergic reaction, dermographism
Urticaria (hives)	Wheals coalesce into extensive reaction; intensely pruritic	—

Lesion	Description / size	Example
Vesicle	Elevated cavity with free fluid up to 1 cm ("blister"); clear serum if wall ruptures	Herpes simplex, early varicella, herpes zoster, contact dermatitis
Bulla	>1 cm; usually single-chambered (unilocular); thin-walled, ruptures easily	Friction blister, pemphigus, burns, contact dermatitis
Cyst	Encapsulated fluid-filled cavity in dermis or subcutaneous layer; tensely elevates skin	Sebaceous cyst, wen
Pustule	Turbid fluid (pus) in the cavity; circumscribed, elevated	Impetigo, acne

[Jarvis pp. 230–231]

Secondary skin lesions (Table 13.4)

Result from change in a primary lesion with time (evolutionary change). Combined designations: papulosquamous, maculopapular, vesiculopustular, papulovesicular. [Jarvis p. 231]

Lesion	Description	Example
Crust	Thickened, dried exudate left when vesicles/pustules burst or dry; red-brown, honey, or yellow	Impetigo (dry, honey-colored), weeping eczematous dermatitis, scab after abrasion
Scale	Compact desiccated flakes, dry or greasy; silvery or white; shed dead keratin	After scarlet fever or drug reaction (laminated sheets), psoriasis (silver, mica-like), seborrheic dermatitis (yellow, greasy), eczema, ichthyosis, dry skin
Fissure	Linear crack with abrupt edges; extends into dermis; dry or moist	Cheilosis (corners of mouth from excess moisture), athlete's foot

Lesion	Description	Example
Erosion	Shallow scooped-out depression; epidermis lost; moist, no bleeding; heals without scar	—
Ulcer	Deeper depression into dermis; irregular; may bleed; leaves scar	Stasis ulcer, pressure injury, chancre
Excoriation	Self-inflicted superficial abrasion; sometimes crusted; from intense itching	Insect bites, scabies, dermatitis, varicella
Scar	Normal tissue lost, replaced by collagen; permanent fibrotic change	Healed surgery/injury, acne
Atrophic scar	Depressed skin level with tissue loss; thinning of epidermis	Striae
Lichenification	Skin thickened by prolonged intense scratching; tightly packed papules; looks like moss	—
Keloid	Benign excess scar tissue beyond original injury; smooth, rubbery, shiny, "clawlike"; smooth and firm; may appear months to years later; most common ages 10–30 years; higher incidence in Black, Latino, and Asian people	Surgery, acne, ear piercing, tattoos, infections, burns

[Jarvis pp. 231–233]

Vascular lesions (Table 13.6) and blanching

Blanching key: stretching the skin between thumb and index finger blanches the normal underlying red tones. Red macules from *dilated blood vessels* blanch momentarily; those from *extravasated blood* (petechiae) do not. Blanching also helps identify a macular rash in dark-skinned people. [Jarvis p. 212]

Lesion	Look / behavior	Key facts
Port-wine stain (nevus flammeus)	Large, flat macular patch on scalp or face, often along cranial nerve V; dark red, bluish or purplish; mature capillaries	Color intensifies with crying, exertion, heat or cold; present at birth and usually does not fade; laser photoablation; early treatment avoids deepening/thickening
"Strawberry mark" (infantile hemangioma)	Raised bright red area, well-defined borders, about 2 to 3 cm; **does not blanch** with pressure	Benign endothelial growth in neonatal period; grows first few months; most involute by age 3 years; main complication is ulceration; oral nonselective beta-blocker treatment
Deep hemangioma (mature)	Reddish-blue, irregularly shaped, solid and spongy vascular mass	May be present at birth, enlarge during first 10 to 15 months; does not involute spontaneously; most need no treatment
Telangiectasia	Permanently enlarged, dilated vessels visible on skin surface	Cosmetics, injection sclerotherapy, or laser
Spider telangiectasia (spider angioma)	Fiery red, star-shaped marking with solid circular center; capillary "legs" radiate from central arterial body	With pressure: central pulsating body and blanching of the extended legs; face, neck, chest; with pregnancy, chronic liver disease, estrogen therapy, or normal. More than 5 occur with significant liver disease
Venous lake	Blue-purple dilation of venules and capillaries in star-shaped, linear, or flaring pattern	Pressure causes them to empty or disappear; lips, ears, face of older adults; electrosurgery or laser

Lesion	Look / behavior	Key facts
Petechiae	Tiny punctate hemorrhages, 1 to 3 mm, round, discrete; dark red, purple or brown; **will not blanch**	Bleeding from superficial capillaries; may indicate abnormal clotting factors; best seen in lighter-melanized areas (abdomen, buttocks, volar forearm); inspect mouth/buccal mucosa and conjunctivae
Purpura	Confluent extensive patch of petechiae and ecchymoses; >3 mm, flat, red to purple macular hemorrhage	Thrombocytopenia, coagulation disorders, scurvy; in old age leaks from capillaries with minor trauma
Ecchymosis	Purplish patch from extravasation of blood into skin; >3 mm diameter	Extravascular red blood cells and pigments in tissue
Contusion (bruise)	Hemorrhage into tissue from mechanical injury (e.g., a blow); skin intact; red-blue/purple immediately or within 24 hours, then blue-purple, blue-green, yellow, brown to disappearing	**Pressure will not cause blanching**; recent bruise in dark skin is deep dark purple; **cannot date a bruise by color**; may also result from bleeding disorders and liver dysfunction
Hematoma	Extravascular presence of red blood cells (with petechiae, ecchymosis)	Listed with erythema from extravascular red blood cells

Purpuric lesions are caused by blood flowing out of breaks in vessels; difficult to see in dark-skinned people. A bruise differs from petechiae, ecchymosis and purpura because those three are not caused by blunt force trauma. [Jarvis pp. 208, 235–237]

> Normal variant: cherry (senile) angiomas are small (1 to 5 mm), smooth, slightly raised bright red dots on the trunk in all adults older than 30 years; they increase in number and size with aging and are not significant. [Jarvis p. 211]

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6. ABCDE melanoma signs and skin cancers

Jarvis summarizes danger signs of pigmented lesions with the mnemonic **ABCDEF**. [Jarvis pp. 207–208]

Letter	Sign	What Jarvis says
A	Asymmetry	Not regularly round or oval; two halves of the lesion look different
B	Border irregularity	Notching, scalloping, ragged edges, poorly defined margins
C	Color variation	Areas of brown, tan, black, blue, red, white, or combination
D	Diameter >6 mm	Size of a pencil eraser — although early melanomas may be diagnosed at a smaller size
E	Elevation or Evolution	Sudden appearance or recent change of a mole
F	Funny looking	"Ugly duckling" sign — the suspicious lesion stands out as looking different from its neighboring nevi

Additional symptoms: rapidly changing lesion; a new pigmented lesion; development of itching, burning, or bleeding in a mole. All of these suggest malignant melanoma and warrant referral. [Jarvis p. 208]

Normal nevi vs. danger

Acquired nevi have symmetry, small size (6 mm or less), smooth borders, and single uniform pigmentation. Junctional nevus is macular only (children/adolescents); progresses to compound nevus (macular and papular) in young adults; intradermal nevus mainly in older age. [Jarvis p. 207]

Malignant skin lesions (Table 13.9)

UV radiation in sunlight and indoor tanning beds promotes all three forms of skin cancer. More than half of a person's lifetime sun damage occurs before adulthood. [Jarvis p. 244]

Cancer	What you see	Key facts
Basal cell carcinoma	Starts as small pink or red papule (may be deeply pigmented) with pearly translucent top and overlying telangiectasia; then rounded pearly borders with central red ulcer, or looks like large open pore with central yellowing	Most common form of skin cancer; slow but inexorable growth; face, ears, scalp, shoulders; low-risk = well-defined borders, higher risk = ill-defined and recurrent; all must be removed
Squamous cell carcinoma	Erythematous scaly patch with sharp margins, 1 cm or more; develops central ulcer and surrounding erythema	Arises from actinic keratoses or de novo; usually hands or head (UV-exposed); less common than basal cell but grows rapidly; lower risk = well defined, primary tumor, no rapid growth; all must be removed
Malignant melanoma	Usually brown; can be tan, black, pink-red, purple, or mixed; often irregular or notched borders; may have scaling, flaking, oozing. Superficial spreading type (most common): brown-to-black macule with irregular borders and color variation. Nodular type: quickly growing blue, black, pink or red nodules, possibly ulceration or bleeding	Malignant transformation of melanocytes, most often cutaneous; about 80% of cutaneous melanomas arise in sun/artificial UV-exposed regions; any body site, most often trunk, head, neck

[Jarvis p. 244]

Risk factors

- Personal or family history of melanoma; atypical, large, or numerous (>50) moles; high UV exposure from sunlight or indoor

tanning beds; persons who sunburn easily or have natural blond or red hair; advancing age (accumulated DNA damage). Most skin cancer deaths are attributable to UV radiation exposure. [Jarvis p. 202]

- UV light damages DNA, causes oncogene mutations and mutation of tumor suppressor gene TP53. [Jarvis p. 202]
- Melanoma is only 1% of skin cancer cases but causes the vast majority of skin cancer deaths and is potentially preventable. White persons: 28 melanoma cases per 100,000; Native American 7; Hispanic 5; Black and Asian 1. Women outnumber men before age 50; by 65 men have double the rates of women, by 80 they are triple. [Jarvis p. 202]
- Indoor tanning increases melanoma risk by 75% if used before age 35 years; melanoma is the number 1 most diagnosed cancer in 25- to 29-year-olds in the United States. [Jarvis p. 202]
- Actinic keratosis is the most frequent premalignant skin lesion in White persons, caused directly by sun/artificial UV radiation; red-tan scaly plaques that become raised and roughened, may develop into squamous cell carcinoma. [Jarvis p. 221]

Health promotion and patient teaching

- Use sunscreen, hats, and long-sleeved tops during outdoor sports and beach or pool activities. [Jarvis p. 223]
- Discourage the "healthy glow" fallacy — sunburns and tans are not healthy; UV rays make skin look wrinkled and older. [Jarvis p. 223]
- Inquire about indoor tanning and discuss its dangers; indoor tanning should NOT be part of anyone's health and fitness routine. [Jarvis p. 223]
- Teach the skin self-examination: well-lighted room, full-length mirror, handheld mirror, family member to check hard-to-see areas (behind ears, back of neck, back); use the ABCDEF rule and report suspicious lesions promptly. [Jarvis p. 223]
- Early detection of any skin lesion increases a healthy outcome. [Jarvis p. 223]

> Jarvis lists EFG nowhere in this chapter — the mnemonic given is ABCDEF (F = "funny looking"/ugly duckling). Do not add EFG unless your instructor's slides include it.

7. Pressure injuries

Pressure injuries (PI) appear on skin over a bony prominence when circulation is impaired, as when confined to bed or immobilized. Immobilization impedes blood delivery (oxygen, nutrients) and venous drainage of metabolic wastes, resulting in ischemia and cell death. Common sites: back (heel, ischium, sacrum, elbow, scapula, vertebra) or side (ankle, knee, hip, rib, shoulder). [Jarvis p. 234]

Risk factors

- Impaired mobility; thin fragile skin of aging; decreased sensory perception; impaired level of consciousness; moisture from urine or stool incontinence, excessive perspiration or wound drainage; shearing injury (being pulled down or across in bed); poor nutrition; infection. [Jarvis p. 234]
- Vasopressor administration (norepinephrine) increases PI risk in critically ill patients because it targets alpha receptors. [Jarvis p. 234]
- Prevention is far more easily accomplished than treatment of existing ulcers. Once stage 3 or 4 ulcers occur, measure wound size daily with disposable rulers (mm and cm), greatest overall length and width. [Jarvis p. 234]
- Aging: thinning skin, loss of collagen, decrease in vascularity and nutrients, loss of subcutaneous cushioning, immobility; delayed wound healing. [Jarvis pp. 201, 222]

Stages of pressure injuries (Table 13.5)

Stage	What you see
Stage 1 — Non-blanchable erythema	Intact skin, red but unbroken. Localized redness in lightly pigmented skin does not blanch with fingertip pressure. Dark skin appears darker but does not blanch. May have changes in sensation, temperature, or firmness.
Stage 2 — Partial-thickness skin loss	Loss of epidermis with exposed dermis. Superficial ulcer looks shallow like an abrasion or open blister with red-pink wound bed. No visible fat or deeper tissue.

Stage	What you see
Stage 3 — Full-thickness skin loss	Extends into subcutaneous tissue, resembles a crater. Subcutaneous fat, granulation tissue, and rolled edges visible, but not muscle, bone, or tendon.
Stage 4 — Full-thickness skin/tissue loss	Involves all skin layers and extends into supporting tissue. Exposes muscle, tendon, or bone; may show slough (stringy matter on wound bed) or eschar (black or brown necrotic tissue), rolled edges, and tunneling.
Unstageable	(Not separately defined in this chapter — full-thickness loss obscured by slough/eschar is described within Stage 4 findings; consult your slides.)
Deep tissue pressure injury (DTPI)	Localized, non-blanchable color change to deep red, maroon, purple in intact or nonintact skin. Dark skin appears darker but does not blanch. Or epidermis may separate revealing dark wound or blood-filled blister. Preceded by pain and temperature change. Begins in muscle closest to bone, in older adults and those with lower BMI, commonly over coccyx, sacrum, buttocks, heels.
PI caused by medical device	PI in pattern or shape of a medical device (O2 tubing, airway pressure masks, IV hub, endotracheal tube, cervical collar, sequential compression devices). Skin PI is staged; mucosal PI cannot be staged.

[Jarvis p. 234]

> Jarvis Ch. 13 does not contain the Braden Scale; check her slides.

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8. Common skin lesions

Children (Table 13.7)

Name	Look	Key fact
Diaper dermatitis	Red, moist, maculopapular patch, poorly defined borders in diaper area along inguinal and gluteal folds	Inflammatory from ammonia, heat, moisture, occlusive diapers; history of infrequent changes

Name	Look	Key fact
Candidiasis	Scalding red, moist patches with sharply demarcated borders, some loose scales	Common cause of diaper dermatitis in infants; aggravated by urine, feces, heat, moisture
Impetigo	Moist thin-roofed vesicles on thin erythematous base; rupture to erosions with thick honey-colored crusts	Highly contagious bacterial infection; most common in infants and children; spreads by direct contact
Atopic dermatitis (eczema)	Recurrent intensely pruritic localized patches; erythematous papules and vesicles with weeping, oozing, flaking, fissures, crusts; erythema in White persons, purplish in Black persons	Chronic inflammatory; overstimulated immune system, strong genetic component, environmental triggers; major quality-of-life effect
Chickenpox (varicella)	Small tight vesicles on trunk then face, arms, legs (not palms/soles); "dewdrop on a rose petal"; erupt in crops, become pustules then crusts; intensely pruritic	Shiny vesicles on erythematous base; successive crops over several days
Measles (rubeola)	Red-purple maculopapular blotchy rash ("coppery") that does not blanch; starts behind ears, spreads over face, neck, trunk, arms, legs; Koplik spots 1 to 3 mm in mouth	Prodrome 2 to 4 days of fever + "three Cs": cough, coryza, conjunctivitis; rash on 3rd or 4th day, cephalocaudal
German measles (rubella)	Pink papular rash, paler than measles; face first, then spreads	Distinguished by neck lymphadenopathy and no Koplik spots; prevented by two spaced doses of MMR vaccine

[Jarvis pp. 238–240]

Adults (Table 13.8)

Name	Look	Key fact
Primary contact dermatitis	Erythema first, then swelling, wheals (urticaria) or maculopapular vesicles, scales; intense pruritus	Local inflammatory reaction to an irritant or allergy; characteristic location gives a clue (example shown: poison ivy)
Allergic drug reaction	Range from mild itching to erythematous symmetric rash to life-threatening anaphylaxis	Immune response requiring prior sensitization; most are mild but the drug must be avoided in future; severe reactions in penicillin allergy
Tinea corporis (ringworm of the body)	Multiple circular lesions with clear centers on chest, abdomen, back of arms; scales hyperpigmented in light skin, depigmented in dark skin	Fungal; characteristic clearing in center
Tinea pedis (athlete's foot)	Small vesicles between toes, sides of feet, soles; grows scaly and hard	Fungal; chronically warm, moist feet — children after gym, athletes, aging adults who cannot dry feet well
Herpes zoster (shingles)	Small grouped vesicles along a cutaneous sensory nerve route, then pustules, then crusts; acute, unilateral, does not cross midline	VZV reactivation of chickenpox virus; commonly on trunk; ophthalmic branch of CN V risks the eye; most common in adults older than 50 years; postherpetic neuralgia in aging adults
Erythema migrans (Lyme disease)	Bull's-eye red macular or papular rash radiating from tick bite (5 cm or larger) with central clearing; regional lymphadenopathy	Spirochete carried by deer tick (Northeast, upper Midwest; May–September); rash in only 50% of cases; fades in 4 weeks; antibiotics shorten symptoms and reduce sequelae

Name	Look	Key fact
Labial herpes simplex (cold sores)	Prodrome of tingling/sensitivity, then tight vesicles, then pustules; acute gingivostomatitis with shallow painful ulcers	HSV; common on upper lip, also oral mucosa and tongue

[Jarvis pp. 241–243]

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9. Hair findings

Normal hair: inspect and palpate

- **Color:** from melanin production, pale blond to total black; graying may begin as early as the 30s from reduced melanin in follicles; onset is genetic [Jarvis p. 213].
- **Texture:** fine to thick, straight/curly/kinky, normally shiny (shine may be lost with dyes, rinses, permanents) [Jarvis p. 213].
- **Distribution:** fine vellus hair over body; coarser terminal hair at eyebrows, lashes, scalp; puberty pattern — pubic first, then axillary, then facial in boys. Female genital pattern = inverted triangle; male = upright triangle with hair up to the umbilicus. Body hair may be diminished in Asians [Jarvis p. 213].
- **Lesions/scalp:** section and lift the hair to view scalp; with itching check behind the ears and occiput. Loose white flakes (seborrhea/dandruff) are common and normal [Jarvis p. 213].
- Hair types: fine vellus hair vs darker, thicker terminal hair (scalp, brows; after puberty axillae, pubic area, male face/chest) [Jarvis p. 200].

Hirsutism

- Excess body hair in females forming a male pattern on the face, chest, and back; indicates increased androgen hormones [Jarvis p. 213].
- Causes: genetics, adrenal or ovarian abnormalities, excess adrenocorticotropic hormones, anabolic steroid ingestion, polycystic ovary syndrome; distribution includes upper lip, face, chest, abdomen, arms, legs [Jarvis p. 247].

- Subjective cue: “unusual hair growth” — hirsutism is shaggy or excessive hair [Jarvis p. 204].

Alopecia (hair loss)

- Alopecia is a *significant* loss of hair; ask gradual vs sudden onset, symmetry, associated fever/illness/stress. Hair loss matters culturally — do not treat as trivial [Jarvis p. 204].
- Loss of eyebrows and scalp hair is expected with chemotherapy and hypothyroidism [Jarvis p. 213].
- Male-pattern balding is a genetic trait: gradual receding of the anterior hairline in a symmetric W shape [Jarvis p. 222]; a symmetric W-shaped balding may occur in the frontal areas [Jarvis p. 201].

Table 13.10 — abnormal conditions of the hair

Condition	Look for	Cause / notes
Toxic alopecia	Patchy, asymmetric balding; growing hairs lost, resting hairs spared	Severe illness or chemotherapy; regrows after illness or drug stopped [Jarvis p. 245]
Tinea capitis (scalp ringworm)	Rounded patchy loss, broken-off hairs, pustules, scales; fluoresces bright blue-green under Wood’s light	Fungal; usually children and farmers; highly contagious from people, animals, soil [Jarvis p. 245]
Traction alopecia	“Fringe sign” marginal loss at frontal/temporal hairline; broken hairs or follicular pustules	Mechanical: tight braids, ponytails, cornrows, weaves, dreadlocks, chemical relaxation; affects one-third of women of African descent [Jarvis p. 245]
Seborrheic dermatitis (cradle cap)	Thick yellow-to-white greasy adherent scales with mild erythema, scalp/forehead; *no* pruritus	Very common in early infancy; differs from eczema by greasy lesions and negative allergy history [Jarvis p. 246]

Condition	Look for	Cause / notes
Folliculitis barbae (“razor bumps”)	Multiple pustules with hair at center on erythematous base, face and neck	Hairs curl back and pierce skin after shaving; common in Black and Latino men with tight curly hair [Jarvis p. 246]
Pediculosis capitis (head lice)	Translucent teardrop nits adherent to shaft near scalp; check behind ears and nape; intense itching	Close head-to-head contact; common in preschool/school age, no socioeconomic correlation [Jarvis p. 246]
Hirsutism	Male-pattern excess hair in a female	Excess circulating androgens [Jarvis p. 247]
Furuncle and abscess	Red, swollen, hard, tender, pus-filled lesion (neck, buttocks)	Usually staphylococcal; furuncle = infected hair follicle, abscess = bacteria introduced by trauma; abscess larger and deeper [Jarvis p. 247]

> Distinguish dandruff (loose white flakes, common/normal) from lice nits (oval, adherent to the hair shaft, intense itching) [Jarvis p. 213].

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10. Nail findings

Normal nails

- Hard plates of keratin; nail plate is clear with fine longitudinal ridges (ridges become prominent with aging) [Jarvis p. 200].
- Pink color comes from the vascular nail bed; the lunula is the white opaque semilunar area over the nail matrix; nail folds overlap posterior and lateral borders; cuticle protects the matrix like a gasket [Jarvis p. 200].
- Shape/contour: surface slightly curved or flat, folds smooth and rounded, edges smooth, rounded, clean [Jarvis p. 213].
- Consistency: smooth, regular, not brittle or splitting; uniform thickness; nail firmly adherent to bed; nail base firm to palpation [Jarvis p. 214].

- Color: translucent plate over an even pink bed; white hairline markings (leukonychia) from trauma/picking are normal in all people; dark-skinned people may have brown-black bands or streaks along the nail edge [Jarvis p. 215].

Profile sign and clubbing

- **Profile sign:** view the index finger in profile; the angle of the nail base should be about **160 degrees**; nail base is firm [Jarvis p. 214].
- **Curved (convex) nails** are a normal variant that may mimic clubbing — but the angle stays 160 degrees or less [Jarvis p. 214].
- **Early clubbing:** angle straightens to **180 degrees**, nail base feels spongy; then the nail becomes convex as the digit grows [Jarvis p. 214].
- **Late clubbing:** inner edge of nail elevates; nail bed angle greater than 180 degrees; distal phalanx rounder, wider, shiny [Jarvis p. 249].
- Causes: congenital cyanotic heart disease, lung cancer, pulmonary diseases [Jarvis p. 214]; chronic lung inflammation, lung cancers, heart defects with right-to-left shunts (growth factor release) [Jarvis p. 249].
- Course: develops slowly over years; can reverse if the primary disease is treated [Jarvis p. 249].

Capillary refill

- With index or middle fingertip at heart level, press the nail edge at least 5 seconds to blanch, release, note color return [Jarvis p. 215].
- Normal return is instant or within a few seconds; a healthy return takes **1 or 2 seconds** [Jarvis p. 215].
- Cyanotic nail beds or sluggish return **>3 seconds** = clinical decline (cardiovascular or respiratory failure, septic shock) [Jarvis p. 215].
- Older adults with prolonged inactivity: cool pale extremities, pale nail beds, prolonged capillary filling time [Jarvis p. 220].

Table 13.11 — abnormal nail conditions (Finding | Look | Cause)

Finding	Look	Cause
Spoon nails (koilonychia)	Concave, spoon-shaped nail	Chronic iron deficiency anemia [Jarvis p. 208]
Beau lines	Transverse furrow/groove across nail, extending to the nail bed; dent starts at cuticle and moves forward	Any trauma temporarily impairing nail formation — acute illness, toxic reaction, local trauma [Jarvis p. 248]
Splinter hemorrhages	Red-brown linear streaks	Damage to nail bed capillaries: vasculitis, trauma or sports injury, endocarditis [Jarvis p. 248]
Paronychia	Red, swollen, tender nail folds; acute = pus in proximal fold, pain, throbbing	Acute usually bacterial; chronic usually fungal from cuticle break in “wet” work [Jarvis p. 248]
Onychomycosis	Color/texture/thickness change, crumbling, plate loosening from distal edge proximally; green where plate separates	Slow persistent fungal infection; fingernails and more often toenails; common in older adults [Jarvis p. 249]
Onycholysis / nail plate separation	Nail plate loosens or separates from bed; green discoloration at separation	Part of onychomycosis pattern in older adults [Jarvis p. 249]
Pitting	Sharply defined pits and crumbling with distal detachment	Often psoriasis; pits/grooves/lines may also indicate nutrient deficiency or acute illness [Jarvis p. 214, 249]

Finding	Look	Cause
Habit-tic dystrophy	Depression down middle of nail or multiple horizontal ridges	Continuous picking of cuticle by another finger injuring nail base/matrix [Jarvis p. 249]
Scabies	Linear or curved elevated burrows in fingers, web spaces, wrists	Scabies mite; intensely pruritic, contagious [Jarvis p. 248]

> Other nail red flags: thickened, ridged nails with arterial insufficiency; spongy nail base with clubbing; jagged, bitten, or traumatized folds suggest nervous picking; chronically dirty nails suggest poor self-care or occupational/cigarette staining [Jarvis p. 213–214]. Suspect melanoma with darkly pigmented nail bands, varied pigmentation, and width >3 mm in light-skinned persons [Jarvis p. 215].

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11. Developmental considerations across the lifespan

Infants

- Hair follicles develop at 3 months' gestation; lanugo covers most skin by midgestation; replaced by vellus hair in first months. Newborn terminal scalp hair may be lost patchily at temples and occiput [Jarvis p. 201, 219].
- Vernix caseosa (thick, cheesy sebum + shed epithelial cells) at birth; green-tinged vernix = meconium staining [Jarvis p. 201, 218].
- Newborn skin thin, smooth, elastic, more permeable → greater risk of fluid loss; sebum present first weeks (milia, cradle cap), then sebaceous glands shrink until puberty; temperature regulation ineffective [Jarvis p. 201].
- Black newborns initially lighter-toned (immature pigment); full melanotic color shows in nail beds and scrotal folds [Jarvis p. 216].

- Mongolian spot: blue-black-to-purple macule at sacrum/buttocks from deep dermal melanocytes, fades in first year; present in 90% of Black, 80% of Asian and Native American, and 9% of White persons — do not confuse with bruising/abuse [Jarvis p. 216].
- Café au lait spot: large round/oval light brown patch, usually present at birth; 6 or more macules each >1.5 cm = neurofibromatosis [Jarvis p. 216].
- Three benign erythematous states: beefy red flush of the first 24 hours; harlequin color change (transient red/pale split); erythema toxicum ("flea bite" rash) in first 3–4 days, no treatment [Jarvis p. 217].
- Two temporary cyanotic states: acrocyanosis (lips, hands, feet; clears with warming) and cutis marmorata (mottling with cool rooms; persistent/pronounced with Down syndrome or prematurity) [Jarvis p. 217].
- Physiologic jaundice after day 3–4 in about half of newborns (RBC hemolysis); jaundice on day 1 = hemolytic disease; after 2 weeks = possible biliary obstruction [Jarvis p. 205, 218].
- Carotenemia: yellow-orange from carotene-rich foods — palms/soles, forehead, nasolabial folds, chin, behind ears, knuckles; *no* sclera/mucosa jaundice; fades in 2–6 weeks [Jarvis p. 218].
- Milia: tiny white papules (sebum plugging) on forehead, eyelids, cheeks, nose, chin — do not squeeze, resolve in a few weeks [Jarvis p. 218].
- Test turgor over the abdomen; poor turgor/tenting = dehydration (with delayed capillary refill, tachypnea) or malnutrition [Jarvis p. 218].
- Nevus simplex (salmon patch/stork bite): flat red-pink patch on face or nape, usually fades in first year [Jarvis p. 218].
- Newborn nail beds may be blue for the first few hours, then pink [Jarvis p. 219].

Children

- Epidermis thickens, toughens, darkens; skin better lubricated; hair growth accelerates; eccrine sweat response to heat is minimal through childhood [Jarvis p. 201].
- Inspect for abuse: cigarette burns, excessive bruising especially above knees or elbows, linear whip marks; history that does not fit

the lesion [Jarvis p. 205].

- Severe or blistering sunburn in childhood increases melanoma risk later [Jarvis p. 205].

Adolescents

- Apocrine glands activate → body odor; sebaceous glands more active → oily skin and acne; subcutaneous fat increases, especially in females [Jarvis p. 201].
- Acne is the most common skin problem of adolescence; almost all teens have some; peak incidence 15–18 years, may start at 7–11 years; face, sometimes chest, back, shoulders [Jarvis p. 219]. Over 85% of teenagers have acne [Jarvis p. 205].
- Secondary sex characteristics: enlarged darkened areolae, breast tissue, coarse pubic then axillary then male facial hair [Jarvis p. 201].
- Darkly pigmented skin: acne may leave hyperpigmentation, keloids, scarring [Jarvis p. 219].

Pregnancy

- Increased metabolism, peripheral vasodilation, increased sweat and sebum; fat deposits on buttocks and hips [Jarvis p. 201].
- Striae: jagged silver-to-pink stretch marks, 2nd trimester, abdomen/breasts/thighs; occur in half of all pregnancies; fade but do not disappear [Jarvis p. 220].
- Linea nigra (brown-black midline abdominal line), chloasma (“mask of pregnancy” facial brown patches; also with oral contraceptives), and spider angiomas from increased estrogen (may resolve after birth) [Jarvis p. 220].
- More than 5 spider angiomas = significant liver disease [Jarvis p. 220].

Older adults

- Skin atrophies: parchment thin, lax, dry, wrinkled by the 70s–80s; loss of elastin, collagen, subcutaneous fat, muscle tone → shearing/tearing injuries [Jarvis p. 201].
- Sweat and sebaceous glands decrease → dry skin (xerosis) and heat-stroke risk; vascularity down, fragility up → senile purpura; wound healing delayed [Jarvis p. 201, 221].

- Hair: fewer melanocytes → gray/white, thin and fine; axillary and pubic hair decreases; unopposed androgens may give women bristly chin/upper-lip hair; men grow coarse terminal ear, nose, brow hairs [Jarvis p. 201, 222].
- Nails: slower growth, lusterless with longitudinal ridges, brittle/peeling/yellowed; thickened misshapen toenails — aging or chronic peripheral vascular disease; fungal infections common [Jarvis p. 202, 222].
- Solar lentigines ("liver spots") in 90% of White persons older than 60; seborrheic keratosis (dark, greasy, "stuck on," benign) vs actinic keratosis (red-tan scaly plaque on sun-exposed skin, premalignant, may become squamous cell carcinoma) [Jarvis p. 220–221].
- Acrochordons (skin tags) on eyelids, cheeks, neck, axillae, trunk; sebaceous hyperplasia = yellow papules with central depression on forehead/nose/cheeks, more common in men [Jarvis p. 221–222].
- Decreased turgor — pinched skin "tents"; handle with care, skin tears from minor trauma [Jarvis p. 222].
- Senile pruritus usually from dry skin, too-frequent bathing, or soaps; older adults may not reach feet for self-care; many cannot date bruising or report pain as "part of growing old" [Jarvis p. 206].

Culture

- Same number of melanocytes in all skin colors; melanin production varies [Jarvis p. 199].
- Common in Black persons: keloids (compact collagen bundles below the epidermis), postinflammatory hypo/hyperpigmentation after acne, pseudofolliculitis ("razor bumps"), melasma; hair shaft elliptical and asymmetric in the follicle → damage from traction/trauma [Jarvis p. 202].
- Lower skin cancer incidence in darkly pigmented Black, Native American, and Asian persons (melanin protects from UV) [Jarvis p. 202].
- Keloids: benign scar overgrowth beyond the wound, smooth rubbery shiny; ear lobes, back of neck, scalp, chest, back; ages 10–30; higher incidence in Black, Latino, Asian people [Jarvis p. 233].

- Dark-skinned people normally have lighter palms, nail beds, and lips [Jarvis p. 207]; a full head of hair equates with vitality in many cultures [Jarvis p. 204].

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12. Documentation (SOAP) examples

Normal sample charting

- **S:** No history of skin disease; no change in pigmentation or nevi; no pruritus, bruising, rash, or lesions. On no medications. No work-related skin hazards. Uses SPF 30 sun-block cream outdoors. [Jarvis p. 224]
- **O:** Skin tan-pink, even pigmentation, no suspicious nevi; warm, dry, smooth, even; turgor good, no lesions. Hair even distribution, thick texture, no lesions or pests. Nails: no clubbing or deformities; beds pink with prompt capillary refill. [Jarvis p. 224]
- **A:** Warm, dry, intact skin. [Jarvis p. 224]

Case 1a — H.H., 3-year-old female

- **S:** Exposed to ill preschool classmate 2 weeks ago; 3 days of fever 101°–102.4° F (38.3°–39° C), fatigue, crankiness; “tiny blisters” on chest/back that crusted; 1 day ago new blisters on shoulders, thighs, face with intense itching. [Jarvis p. 224]
- **O:** T 101° F (38.3° C), BP 100/63 (sitting), P 100, R 24; fatigued, irritable. Generalized vesiculopustular rash on face, trunk, upper arms, thighs: small vesicles on face; pustules and red-honey crusts/scabs on trunk; skin warm, dry, good turgor. TMs pearly gray; tonsils 1+ without exudate; no lymphadenopathy; heart and lungs clear. [Jarvis p. 224]
- **A:** Varicella; acute pain and pruritus; potential for transmission; lack of information on vaccine benefits. [Jarvis p. 224]

Case 2 — B.G., 79-year-old, s/p hip replacement

- **S:** Aching pain in left hip (nonoperative side). [Jarvis p. 225]
- **O:** Erosion 2 × 2 cm with surrounding erythesis over left ischium; moist, no active bleeding; area very warm and tender. [Jarvis p. 225]

- **A:** Pressure injury, left hip; acute pain; decreased mobility. [Jarvis p. 225]

Case 3 — M.G., 62-year-old female

- **S:** Tingling, itching, severe non-radiating pain on right flank × 4 days; "weird rash" appeared this morning. Multivitamin only. [Jarvis p. 225]
- **O:** T 98.4° F (36.9° C), P 89, R 18, BP 116/72. Zosteriform rash on right flank ~7 cm long; some vesicles intact, others eroded from scratching; surrounding skin red; left flank intact. [Jarvis p. 225]
- **A:** Herpes zoster, right flank; acute pain; potential for infection R/T broken vesicles. [Jarvis p. 225]

> Jarvis notes that a detailed plan is not given in the text — develop your own treatment plan as a critical-thinking exercise [Jarvis p. 224].

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13. Quick-review tables

Key numbers

Number	What it means	Page
160 degrees	Normal nail base angle (profile sign); 160 or less = normal curved nail	[Jarvis p. 214]
180 degrees / >180	Early clubbing / late clubbing with spongy or elevated base	[Jarvis p. 214, 249]
1–2 seconds	Healthy capillary refill (press ≥5 seconds at heart level)	[Jarvis p. 215]
>3 seconds	Sluggish refill = clinical decline (CV/respiratory failure, septic shock)	[Jarvis p. 215]
<1 cm vs >1 cm	Macule vs patch; papule vs plaque; vesicle vs bulla	[Jarvis p. 230]
1–3 mm	Petechiae (tiny punctate, non-blanching); purpura >3 mm	[Jarvis p. 237]
1–5 mm, age >30	Cherry (senile) angiomas on the trunk, increase with age	[Jarvis p. 211]

Number	What it means	Page
6 mm	Diameter danger sign for pigmented lesions (pencil eraser)	[Jarvis p. 208]
3–4 seconds (thumbs)	Press thumbs for edema check at ankle malleolus/tibia	[Jarvis p. 210]

More key numbers

Number	Fact	Page
20 sq ft	Skin surface area in the average adult; epidermis replaced every 4 weeks	[Jarvis p. 199]
~500 mL/day	Insensible perspiration fluid loss	[Jarvis p. 201]
90% / 80% / 9%	Mongolian spots: Black / Asian & Native American / White persons	[Jarvis p. 216]
6 or more, >1.5 cm	Café au lait macules diagnostic of neurofibromatosis	[Jarvis p. 216]
Day 3–4 vs day 1	Physiologic jaundice vs pathologic (hemolytic disease)	[Jarvis p. 205, 218]
>5	Spider angiomas suggesting significant liver disease	[Jarvis p. 220]
75% / before age 35	Increased melanoma risk from indoor tanning	[Jarvis p. 202]

Look-alike pairs

Pair	How to tell them apart
Curved nail vs clubbing	Curved = angle 160 degrees or less, base firm; clubbing = 180 degrees or more, spongy base [Jarvis p. 214]
Carotenemia vs jaundice	Carotenemia = yellow-orange palms/soles/forehead but *no* sclera or mucosa yellowing; jaundice = sclera and hard/soft palate junction first [Jarvis p. 209, 218]

Pair	How to tell them apart
Dandruff vs lice nits	Loose white flakes (normal) vs oval nits adherent to the shaft with intense itching [Jarvis p. 213]
Mongolian spot vs bruise	Blue-black macule at sacrum/buttocks, present at birth, fades in year 1 [Jarvis p. 216]
Bruise vs petechiae/purpura/ecchymosis	Bruise follows blunt trauma; the other three are not caused by blunt force; none blanch [Jarvis p. 237]
Measles vs rubella	Rubella is paler, appears on face first, has neck lymphadenopathy and *no* Koplik spots [Jarvis p. 240]
Cradle cap vs eczema	Cradle cap = greasy yellow-pink scales, *no* pruritus, negative allergy history [Jarvis p. 246]
Herpes zoster vs genital herpes	Zoster = linear, unilateral, does not cross midline [Jarvis p. 243]

Where to look in dark skin

Color change	Best sites in dark skin
General / pallor / cyanosis / jaundice	Conjunctivae, oral mucosa, under the tongue, buccal mucosa, sclera (hard–soft palate junction for jaundice) [Jarvis p. 208–209, 226–227]
Erythema	Cannot see it — palpate for warmth, taut skin, deep hardening [Jarvis p. 208, 227]
Petechiae	Lighter melanized areas: abdomen, buttocks, volar forearm; also mouth and conjunctivae [Jarvis p. 237]
Macular rash	Stretch/blanch the skin between thumb and finger for contrast [Jarvis p. 212]
Mongolian spot	Sacrum/buttocks — normal, not abuse [Jarvis p. 216]

14. Likely test traps

- Normal nail base angle is **160 degrees**, not 180; 180 degrees means *early clubbing* [Jarvis p. 214].
- A convex/curved nail is a normal variant — the angle stays 160 degrees or less and the base is firm [Jarvis p. 214].
- Clubbing: nail base feels **spongy** early; late clubbing is angle **greater than 180 degrees** with a rounder, wider, shiny distal phalanx [Jarvis p. 214, 249].
- Clubbing can **reverse** if the primary disease is treated [Jarvis p. 249].
- Capillary refill: fingertip at **heart level**, press **at least 5 seconds**, normal return 1–2 seconds; >3 seconds signals decline [Jarvis p. 215].
- Beau lines are **transverse** grooves that start at the cuticle and move forward as the nail grows [Jarvis p. 248].
- Splinter hemorrhages = red-brown linear streaks from capillary damage; think endocarditis, vasculitis, or trauma [Jarvis p. 248].
- Spoon nails (koilonychia) point to chronic **iron deficiency anemia** [Jarvis p. 208].
- Acute paronychia is usually bacterial; chronic is usually fungal and linked to “wet” work [Jarvis p. 248].
- Onychomycosis begins at the **distal** edge and progresses proximally; more common in toenails and older adults [Jarvis p. 249].
- Tinea capitis fluoresces **bright blue-green** under Wood’s light; a Wood’s light is filtered UV used in a darkened room [Jarvis p. 212, 245].
- Traction alopecia shows the “fringe sign” at the frontal/temporal hairline and affects one-third of women of African descent [Jarvis p. 245].
- Hirsutism = male-pattern excess hair in females from excess **androgens**; absent/sparse genital hair suggests endocrine abnormality [Jarvis p. 213].
- Cradle cap has **no pruritus**; eczema is intensely pruritic [Jarvis p. 246].
- Physiologic jaundice = day 3–4; day 1 jaundice = hemolytic disease; jaundice after 2 weeks = biliary obstruction [Jarvis p.

205, 218].

- Carotenemia spares the sclera and mucous membranes; jaundice shows first at the hard/soft palate junction and sclera [Jarvis p. 209, 218].
- Pallor in dark skin = ashen gray or yellowish brown with loss of the normal luster; check conjunctivae near the canthi and nail beds [Jarvis p. 208].
- An anemic person may be hypoxemic **without** looking blue; a polycythemic person looks ruddy blue at all times and may not be hypoxemic [Jarvis p. 209].
- Petechiae and purpura do **not** blanch; red macules from dilated vessels do blanch [Jarvis p. 212, 237].
- Cherry angiomas are normal in all adults older than 30 and increase with age — not significant [Jarvis p. 211].
- Edema makes dark skin look **lighter** and masks jaundice/cyanosis; unilateral edema = local cause, bilateral/generalized = central (heart or kidney failure) [Jarvis p. 210].

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15. Module 3 vocabulary

#	Term	Meaning
1	**alopecia**	Hair loss, patchy or all over (e.g., chemotherapy, fungal infection, autoimmune disease). [Jarvis p. 213, 222]
2	**bulla**	Raised, fluid-filled blister larger than 1 cm (e.g., burn blister, pemphigus). [Jarvis p. 230]
3	**capillary refill**	How fast color returns after pressing a nail bed and releasing; normal under 2 seconds (Jarvis: 1 to 2). Slower refill = poor peripheral circulation. [Jarvis p. 215]
4	**cherry angioma**	Small, smooth, bright red raised dot on the trunk; common, harmless, increases with age (after about 30). [Jarvis p. 211]

#	Term	Meaning
5	**clubbing**	Angle between nail and nail base reaches 180 degrees or more with a swollen fingertip; from long-term lack of oxygen (chronic heart or lung disease). Normal angle about 160 degrees. [Jarvis p. 214]
6	**comedo**	Plugged hair follicle (pore) filled with oil and keratin; open = blackhead, closed = whitehead; first lesion of acne. [Jarvis p. 219]
7	**confluent**	Lesions that run together and merge so you cannot see where one ends (e.g., hives). [Jarvis p. 228]
8	**cyanosis**	Bluish skin and mucous membrane color from too little oxygen in the blood; in dark skin check lips, tongue, conjunctiva, palms, soles. [Jarvis p. 208]
9	**cyst**	Encapsulated, raised lump of fluid or semisolid material palpable in the dermis or under the skin (e.g., sebaceous cyst). [Jarvis p. 231]
10	**diaphoresis**	Heavy, profuse sweating (e.g., fever, exercise, anxiety, shock, heart attack). [Jarvis p. 210]
11	**erythema**	Skin redness from increased blood flow (inflammation, fever, sunburn, blushing); in dark skin feel for warmth or swelling since redness is hard to see. [Jarvis p. 208]
12	**hematoma**	Collection of blood trapped under the skin from a broken vessel, making a raised bruise-like lump. [Jarvis p. 208]
13	**homogeneous**	Even and the same all over (uniform skin color, or a lesion with even color throughout). [Jarvis p. 207]
14	**jaundice**	Yellow skin, sclera, and mucous membranes from high bilirubin (liver disease or red cell breakdown); check the sclera first, especially in dark skin. [Jarvis p. 209]
15	**keloid**	Overgrown raised scar spreading past the original wound edges; more common in people with darker skin. [Jarvis p. 233]

#	Term	Meaning
16	**macule**	Flat spot with color change only, under 1 cm, not palpable (e.g., freckle, flat mole); flat and larger than 1 cm is a patch. [Jarvis p. 230]
17	**nevus (mole)**	Cluster of pigment cells, flat or raised, tan to brown; screen with ABCDE (Asymmetry, Border, Color, Diameter over 6 mm, Evolving) for melanoma. [Jarvis p. 207]
18	**pallor**	Paleness from less blood flow or low hemoglobin (anemia, shock, cold); in dark skin ashen gray or yellowish brown — check conjunctiva, lips, nail beds. [Jarvis p. 208]
19	**petechiae**	Tiny flat red or purple dots (under 2 mm) from bleeding under the skin; do not blanch; may mean a bleeding or clotting problem. [Jarvis p. 212, 237]
20	**primary lesion**	Lesion arising directly from a disease process on previously normal skin (e.g., macule, papule, vesicle, wheal, nodule). [Jarvis p. 211, 230]
21	**purpura**	Flat red-purple patches larger than petechiae (over 3 mm) from bleeding under the skin; do not blanch; seen with low platelets or blood vessel problems. [Jarvis p. 237]
22	**secondary lesion**	Change growing out of a primary lesion over time or from scratching, healing, or infection (e.g., crust, scale, scar, ulcer, fissure, excoriation, lichenification). [Jarvis p. 211, 231]
23	**skin tags**	Soft small flaps of skin hanging off a stalk, common on neck, armpits, eyelids; harmless. [Jarvis p. 221]
24	**skin tenting**	Pinched skin stays up in a ridge instead of snapping back — poor turgor, usually dehydration. [Jarvis p. 210, 222]
25	**spider angioma**	Bright red center with thin red lines like spider legs, usually face, neck, arms, upper chest; blanches with pressure; in liver disease, pregnancy, vitamin B deficiency, sometimes healthy people. [Jarvis p. 220, 236]

#	Term	Meaning
26	**striae**	Silvery or pink stretch marks from rapid skin stretching (pregnancy, weight gain, growth); purple-blue ones can point to Cushing syndrome. [Jarvis p. 220]
27	**turgor**	Skin elasticity, tested by pinching skin under the collarbone and watching return; normal snaps back at once, slow return means dehydration. [Jarvis p. 210]
28	**vesicle**	Raised, fluid-filled blister under 1 cm (e.g., chickenpox, herpes, shingles). [Jarvis p. 230]
29	**vitiligo**	Patches losing all pigment and turning white as pigment cells are destroyed (thought autoimmune); more noticeable on dark skin. [Jarvis p. 207]
30	**wheal**	Raised, red, itchy, irregular area of skin swelling that comes and goes quickly (e.g., hives, mosquito bite, allergic reaction). [Jarvis p. 230]
31	**zosteriform**	Lesions arranged in a band along a nerve path (dermatome), usually one-sided, like shingles (herpes zoster). [Jarvis p. 229]